Daily Study Log: Exploratory Data Analysis (EDA)

Date: March 17, 2026

Topic: End-to-End EDA and Data Preprocessing on COVID-19 Dataset

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1. Objective

The goal of today's session was to perform a comprehensive Exploratory Data Analysis (EDA) on a COVID-19 dataset, focusing on data cleaning, feature engineering, visualization, and preparing the data for machine learning through scaling and dimensionality reduction.

2. Key Activities & Implementation

- **Data Cleaning:** * Identified missing values using isnull().sum() and visualized them with a Seaborn heatmap.
    - Handled null values by imputing the **mean** for numerical columns (*Confirmed, Deaths, Recovered*) and labeling missing categorical data (*Province/State*) as "Unknown."
- **Feature Engineering:** * Created a new feature, ActiveCases, by calculating:

$$ActiveCases = Confirmed - Deaths - Recovered$$

    - Engineered a binary target variable, CaseCategory, based on the median of confirmed cases.
    - Extracted temporal features (*Year, Month*) from the ObservationDate after converting it to a datetime object.
- **Statistical Analysis & Outlier Handling:**
    - Generated distribution plots and boxplots to identify skewness and extreme values.
    - Applied the **Interquartile Range (IQR) Method** to filter out outliers in the Confirmed cases column to ensure model stability.
- **Data Visualization:**
    - **Correlation Heatmap:** Analyzed the linear relationships between numerical features.
    - **Time Series Trend:** Plotted monthly confirmed cases to visualize the pandemic's progression.

    - **Geographical Analysis:** Visualized the top 10 regions with the highest case counts using bar charts.

- **Advanced Preprocessing:**
    - **Label Encoding:** Converted categorical regions into numerical formats.
    - **Standardization:** Used StandardScaler to bring features to a common scale ($mean=0, std=1$).
    - **Dimensionality Reduction (PCA):** Implemented Principal Component Analysis to retain **95%** of the variance while reducing feature redundancy.
    - **Handling Class Imbalance:** Applied **SMOTE** (Synthetic Minority Over-sampling Technique) to balance the CaseCategory target classes.

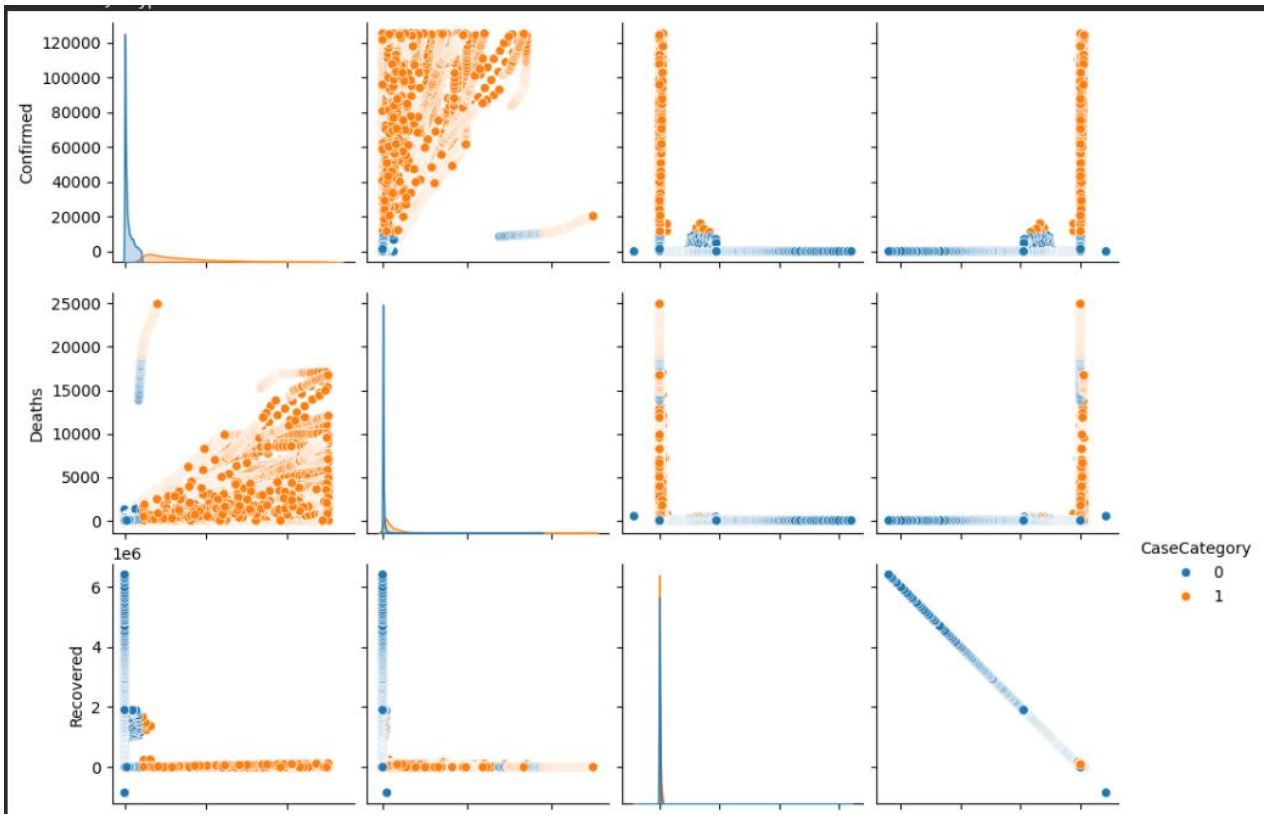
3. Key Insights

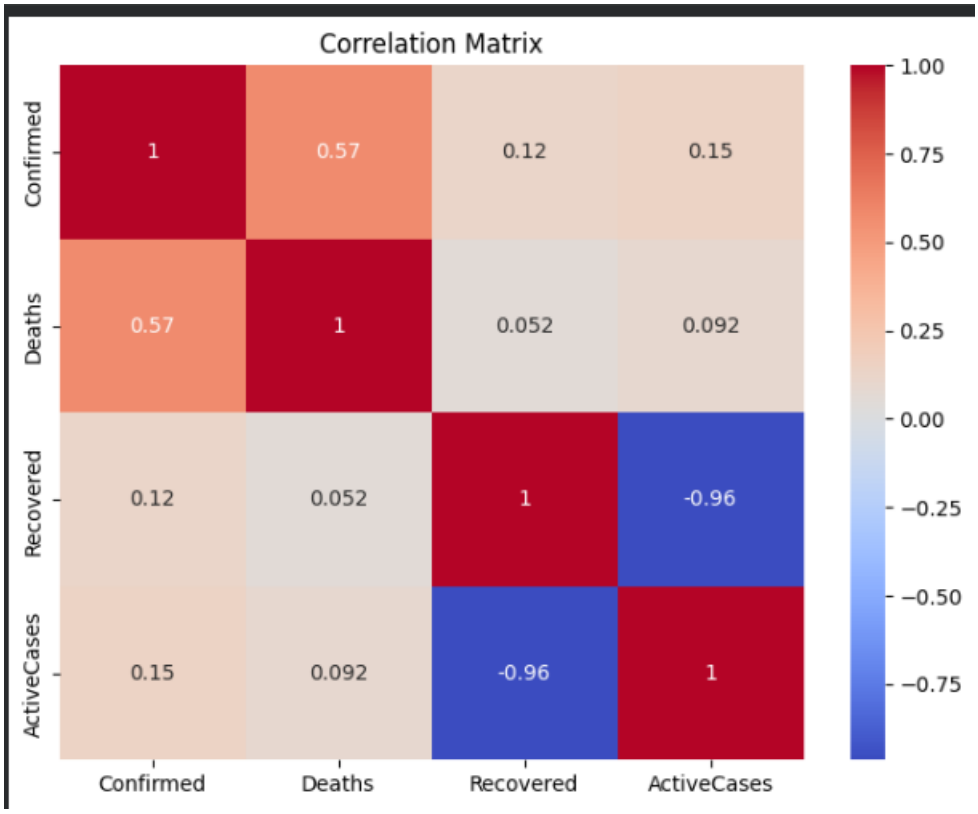
- The data showed a high correlation between *Confirmed* cases and *Deaths*, as expected.
- Outlier removal significantly reduced the dataset noise, preventing extreme values from biasing the statistical summary.
- PCA effectively reduced the feature space, which will optimize future model training time without significant loss of information.

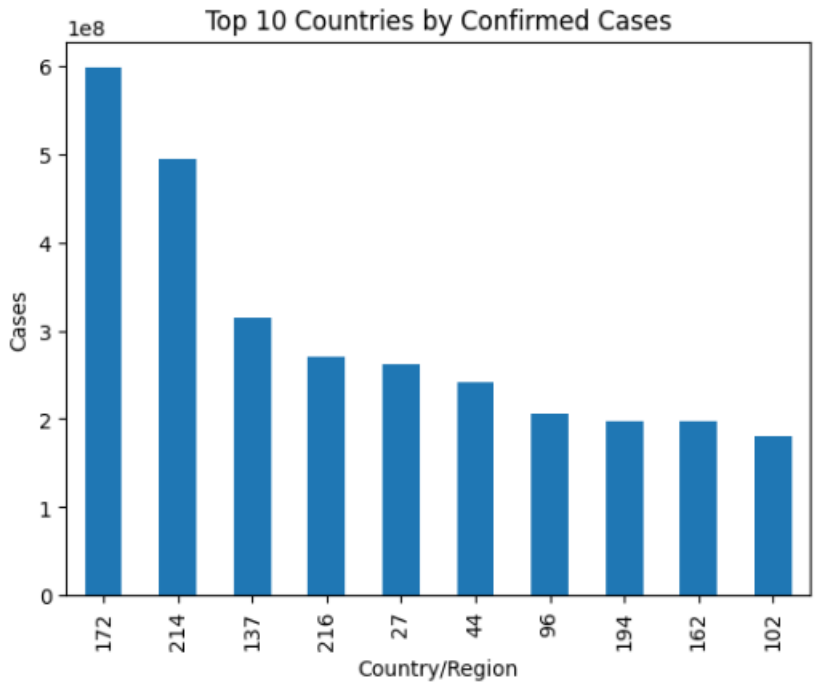
4. Libraries Used

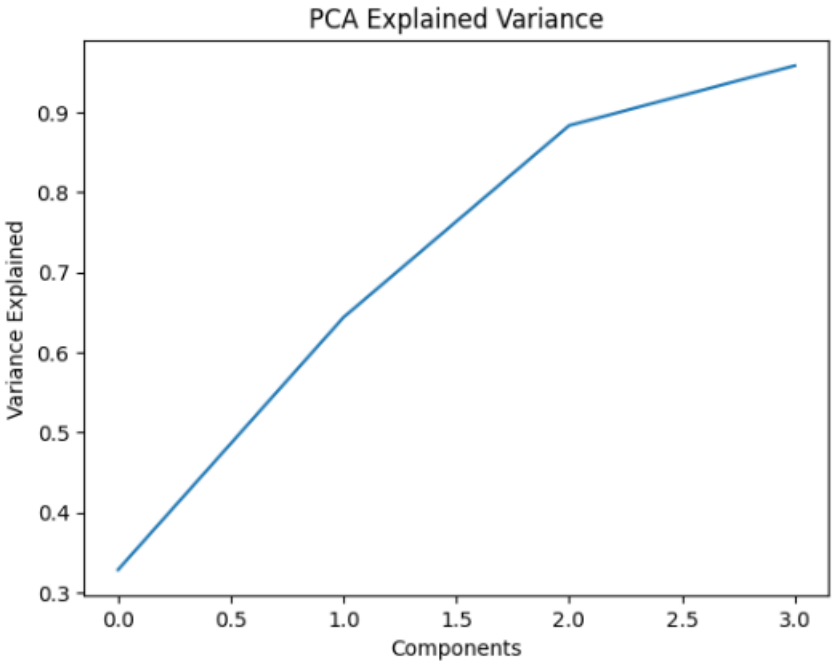
- **Data Prep:** pandas, numpy
- **Visualization:** matplotlib, seaborn
- **Machine Learning:** scikit-learn (PCA, Scaling, Encoding), imblearn (SMOTE)

CODE OUTPUT


Confirmed
Deaths
Recovered
1e6
CaseCategory
0
1


Correlation Matrix
Confirmed
Deaths
Recovered
ActiveCases
1
0.57
0.12
0.15
0.57
1
0.052
0.092
0.12
0.052
1
-0.96
0.15
0.092
-0.96
1
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75


Top 10 Countries by Confirmed Cases
1e8
6
5
4
3
2
1
0
Cases
172
214
137
216
27
44
96
194
162
102
Country/Region


PCA Explained Variance
0.9
0.8
0.7
0.6
0.5
0.4
0.3
Variance Explained
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Components